

DIVYANSHU GOYAL

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SUMMARY

Applied Scientist and multimodal researcher with 10 years at the intersection of frontier ML research and production engineering. I lead the training and fine-tuning of **vision-language models** at scale, design novel **evaluation and alignment** methods (**5 patents**, peer-reviewed and arXiv publications), and previously led the AI powering **Adobe GenStudio for Performance Marketing** (now serving Fortune 500 clients including Microsoft and AT&T). Deep expertise across **VLM training**, **multimodal reasoning**, and **LLM pretraining & scaling laws**, paired with the systems engineering to deploy them in production.

EXPERIENCE

Applied Scientist – Adobe Unified Platform, San Jose

June 2021 – Present

♦ *Vision-Language Model Training & Evaluation*

- Fine-tuned a **Gemma-3-12B vision-language model** with **LoRA** across **8× H100 GPUs** (DDP, bfloat16, Flash Attention 2 + Liger kernels) for large-scale marketing-quality assessment.
- Pioneered an **observable injection** technique that grounds VLM quality reasoning in **14+ computed pixel-level metrics** (sharpness, exposure, color balance); built an interpretable scorer over these signals reaching **Pearson $r \approx 0.88$** at **MAE 0.77**, outperforming GPT-5.4 (MAE 1.34) and Claude Opus 4.7 (MAE 1.95) baselines.
- Developed the **MQCore evaluation benchmark** (inspired by DCLM CORE) and a fault-tolerant async annotation pipeline processing **100+ images/min** over the **700K-image KADIS** corpus.

♦ *Multimodal Research & Benchmarking*

- Authored **DistortBench** (CVPR FGVC13 Workshop 2026): a **13.5K-question** diagnostic benchmark spanning **27 distortion types**, 6 perceptual categories, and 5 severity levels.
- Evaluated **18+ frontier VLMs** (GPT-5.4, Claude Sonnet 4.6, Qwen, InternVL, Gemma) with bootstrap confidence intervals and balanced-accuracy metrics, exposing low-level perception as a major weakness: top model **61.9%** vs. **65.7% human baseline**.
- Created a curiosity-driven **LLM-as-a-judge** framework for personalized creative evaluation using Bayesian surprise as a quantifiable novelty signal (arXiv 2025), achieving **23% F1** and **38% correlation** improvements over baselines.

♦ *Production Multimodal Systems & Agents*

- Architected the **Unified Brand Service** (90K+ LOC across 3 global regions) powering **Adobe GenStudio**: multimodal brand-DNA extraction, RAG-based retrieval, and GPT-4-Vision content validation.
- Built a **multimodal agent** that drives Adobe Photoshop from natural language via a **49-tool** observe-think-act loop with vision grounding and an iterative quality-critic refinement layer.
- Built the founding prototype of **Adobe GenStudio for Performance Marketing** (launched Oct 2024, now serving Fortune 500 clients including Microsoft and AT&T); invented patented VLM pipelines for **cross-cultural image adaptation** and **constrained machine translation** (US Patents 2026).

Machine Learning Engineer – Adobe Inc., Bangalore

July 2016 – August 2019

- Co-engineered Adobe's production **ML inference platform**, sustaining **3,500 QPS/node** at **0.3–0.6ms p99** latency through flame-graph profiling, JMH micro-benchmarking, and lock-free design.
- Architected a **GC-free asynchronous event pipeline** on the LMAX Disruptor ring buffer, decoupling telemetry from prediction to eliminate back-pressure at peak load.
- Built an end-to-end **model lifecycle platform**: versioned deployment, live monitoring, and staged rollout across dev, staging, and production.
- Co-authored the [Adobe Tech Blog](#) on engineering high-throughput, low-latency ML services.

PUBLICATIONS & PATENTS

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment

V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari

A curiosity-driven framework that personalizes creative-writing evaluation to individual judgments via LLM-as-a-judge, targeting the subjectivity gap in LLM assessment.

Cultural Adaptation of Images Using Generative AI

A. Eppa, M. Anand, **D. Goyal**

Multi-stage pipeline where a vision-language model grounds source images as text, conditioned on region-specific cultural guidelines, and a text-to-image model renders culturally adapted output.

Guiding Language Translation with Translation Documents Using ML

D. Goyal, A. Eppa, M. Anand

Structured approach to constrained machine translation by extracting guidelines into language-agnostic facets and computing facet-specific adherence scores for linguistic and stylistic alignment.

Parameter-Efficient Model Serving Systems

Y. Wang, N. Vangala, M. Anand, **D. Goyal**, et al.

System for efficiently serving fine-tuned models by dynamically loading parameter-efficient layers into pre-loaded base models, minimizing network transfer costs and container startup time.

Brand-Aligned Content Generation with Structured Knowledge

D. Goyal, M. Anand, J. Mathew

Generative system that transforms unstructured brand information into structured brand DNA elements with confidence scoring, enabling brand-aligned content creation across channels.

SELECTED PROJECTS & TECHNICAL WRITING

VibeNanoChat:GPT-2 Scale LLM Pretraining Framework

From-scratch decoder-only transformer (RoPE, RMSNorm, sliding-window attention) with one-knob depth-parameterized scaling, a custom distributed **Muon** optimizer with ZeRO-2 state sharding, and Flash-Attention-3 / bfloat16 training on 8x H100. Empirically verified the **C = 6ND** compute law across 40+ scaling-law runs.

A Lazy Engineer’s Guide to Scaling Transformers

Deriving all transformer architecture dimensions, training hyperparameters, and compute schedules from a single integer (depth), enabling trivial multi-scale training sweeps.

Going Fast: Every Optimization That Made LLM Training Fly

Six concrete optimizations (TF32, BF16 mixed precision, torch.compile, Flash Attention, parallel DataLoaders, and pinned memory) that collectively drove significant MFU improvements during GPT-2 scale training.

TECHNICAL SKILLS

Research:	LLM Pretraining & Alignment (SFT, RLHF, DPO), Vision-Language Models, Multimodal Agents, Evaluation & Benchmarking, Data Curation
Training & Serving:	PyTorch, Hugging Face Transformers/PEFT, DeepSpeed, FSDP, vLLM, Flash Attention, LoRA/QLoRA, CUDA, Weights & Biases
Infrastructure:	Docker, LangChain, RAG pipelines
Programming:	Python, C++, JavaScript

EDUCATION

Georgia Institute of Technology – M.S. Computer Science (Machine Learning)

2019 – 2021 | **GPA: 4.0**

BITS Pilani – B.E. Computer Science, M.Sc. Mathematics

2011 – 2016 | **GPA: 8.75/10**